Sub Code:NCE702

Paper Id: 100702 Roll No:

B. TECH
(SEM -VII) THEORY EXAMINATION 2019-20
WATER RESOURCES ENGINEERING

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

a. Define Furrow and Drip irrigation system.
b. Write down the relation between Duty and Delta.
c. Define Outlet Factor.
d. What is Evapotranspiration?
e. Define intensity of irrigation.
f. Explain Lacey's silt factor.
g. What do you meant by a "Cross-Drainage Works"?
h. Write a short note on Canal siphon and Aqueduct?
i. Define silting and scouring in canals.
j. What is water logging?

SECTION B

2. Attempt any *three* of the following: 10x3=30

a. Describe the concept of hydrologic cycle with the help ofa neat sketch. What are the different components ofthe hydrologic cycle? Write down water budget equation for surface flow.
b. Write a short note on 'synthetic Unit Hydrograph. Howwill you derive the synthetic unit hydrograph from a numberof unit hydrograph? Illustrate the method with suitableexample in a tabular form.
c. What is the problem of water logging? What are the poor effects of water logging? Describe some suitable remedial measures against water logging in brief.
d. What do you mean by river training works? Describe the various method used for river training work.
e. Define the following terms in brief:
   i. Well losses
   ii. Specific Capacity
   iii. Specific yields
   iv. Well efficiency

SECTION C

3. Attempt any *one* part of the following: 10x1=10

a. Write short notes on:
   i. Intensity - Duration Curve and
   ii. Probabilistic Maximum Precipitation Curve.
b. Define surface runoff. Explain the factors affecting the runoff.

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4. Attempt any *one* part of the following: 10x1=10

a. Describe the various method of irrigation system. Define sprinkler irrigation system with neat sketch.

b. What is meant by crop rotation? What are theadvantages of crop rotation? Describe in briefwithsuitable examples.

5. Attempt any *one* part of the following: 10x1=10

a. Using Lacey's theory, design a trapezoidal irrigation channel (side slope, 1H: 2V) carrying discharge of 40 m^3/sec. Take silt factor as 1.0.

b. What do you understand by regime channel? Explain the initial regime and final regime of a channel in Lacey's theory.

6. Attempt any *one* part of the following: 10x1=10

a. What is cross drainage works? What are the various types of cross drainage works?

b. Design a concrete lined channel to triangular section to carry a discharge of 45 m^3/secat a slope of 1 in 1000. The side slopes of the channel are 1.5:1 and Manning's rugosity coefficient for lining material as 0.018.

7. Attempt any *one* part of the following: 10x1=10

a. Describe Confined and Unconfined aquifer with suitable diagram. Derive the expression for the discharge through confined aquifer.

b. Write short notes on :

i. Well shrouding and well development

ii. Types of open wells

iii. Infiltration galleries.

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